ClimaWatch: locally calibrating a global PM2.5 grid in Chișinău and București — a pre-registered benchmark and cost model

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1. Aim and hypothesis

Free global air-quality grids are easy to embed in applications but coarse: they miss local urban behaviour. We tested whether a small model that uses only information available at forecast time can correct the PM2.5 field of the OpenWeatherMap (OWM) air-pollution product toward what local sensors measure. Hypotheses were fixed before any model was run (repository file *research/PREREGISTRATION.md*). **H1:** gradient-boosted trees (GBT) reduce hourly PM2.5 RMSE by at least 22% against the raw grid over N ≥ 100 held-out 24-hour windows. **H2:** the same model reduces the miss rate of unhealthy-air alerts (daily mean ≥ 15 µg/m³, WHO 2021 guideline) by at least 18%. A hypothesis counts as supported only if the point estimate reaches the target and the 95% bootstrap interval excludes zero.

2. Methodology

Data (1 Jan–31 Oct 2026). Ground truth: hourly median of the Sensor.Community low-cost particle sensors within 12 km of each centre (5 in Chișinău, 10 in București; plausibility filter; at least 2 and 3 sensors per hour). Baseline input: OWM hourly air-pollution history. Weather: ERA5 reanalysis. Twenty features (grid pollutants, temperature, humidity, wind, pressure, precipitation, boundary-layer height, hour, weekday); no sensor value is an input. **Models:** raw grid; scale + offset; ridge regression; GBT (250 trees, depth 4), implemented from scratch in TypeScript with hyper-parameters fixed in advance. **Validation:** blocked 6-fold cross-validation per city with 2-day gaps, so every one of the 576 city-days (13,791 hours) is tested exactly once; 2,000-resample day-block bootstrap for intervals. **Stated limits:** OWM keeps no forecast archive, so its value for the target hour stands in for the forecast (perfect-prognosis assumption): the experiment tests the calibration step, not forecast skill. The sensors are low-cost and read high in humid air, so a humidity < 75% check was added.

3. Quantitative benchmark results

Table 1. Hourly PM2.5 error and alerts at ≥ 15 µg/m³ (576 windows, 49 unhealthy days)

Method	RMSE	Change	MAE	r	Missed alerts	F1
Raw OWM grid	6.28	—	3.97	0.49	31 (63%)	0.39
Scale + offset	5.52	−12.0%	3.76	0.49	36 (74%)	0.37
Ridge regression	5.00	−20.3%	3.24	0.61	33 (67%)	0.42
Gradient-boosted trees	4.99	−20.5%	3.15	0.65	23 (47%)	0.52

H1 is partially supported: GBT lowered RMSE by 20.5% (95% interval 14.5–26.4%), which is 1.5 points short of the 22% target, although the interval excludes zero. **H2 is not supported:** the miss rate fell by 25.8% (63% → 47%), above the 18% target, but with only 49 unhealthy days the interval is 0–48.6% and does not exclude zero. The gain held in the heating season (−24.2%) and on dry hours only (−27.2%), and differed by city (Chișinău −39%, București −11%). GBT did not beat the linear model (−20.5% vs −20.3%) and generalised worse to months it had not seen (−2.7% vs −16.2%); inference takes 10.7 µs per prediction for GBT (0.24 ms per 24-hour window) and 0.035 µs for ridge.

4. Economic and social applicability

Running cost (model built from the deployed code). One visit causes about 30 OWM calls (26 for the world globe). For 10,000 users with 4 visits a month, an uncached service makes 1.2 million calls, above the free quota of 1 million; CDN caching reduces this to 239,000 (−80.1%; −92.6% at 100,000 users) and raises the free-tier capacity from 8,333 to 119,706 users a month. Hosting and API cost is therefore €0 a month and hardware €0 (any browser). Covering 100 sites with dedicated display monitors would cost €31,900 (IQAir AirVisual Pro, €319 list) or US$20,900–28,900 (PurpleAir Touch/Flex); these measure one spot directly, which ClimaWatch does not. **Health effect**

(order of magnitude). With relative risks of 1.028–1.048 per 10 μg/m³ [1, 2], 14–35% behavioural reductions in outdoor activity on alert days [3, 4] and €215–€1,042 per emergency episode [5, 6], moving outdoor time to the cleanest hours avoids 0.001–0.63 emergency asthma episodes a year per 10,000 users (€0.2–€655; base case €6). The baseline episode rate is an assumption. Social value therefore lies in free, personalised hourly advice, not in large healthcare savings.

5. Conclusions

Local calibration of a free global grid reduces PM2.5 error by about one fifth in two cities, including in the heating season, but the pre-registered accuracy target was narrowly missed and the alert benefit remains unproven. A per-city ridge model is the best deployment candidate: as accurate as trees, more robust, and about 300 times cheaper. Next steps are logging live OWM forecasts to remove the perfect-prognosis assumption, adding reference-grade stations and more cities to obtain enough unhealthy days, and replacing the baseline episode rate with national statistics.

References

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Availability. Code, hourly datasets, pre-registration and logbook are in the project repository (folders *research/* and *docs/moldsef/*); every number above is produced by *research/benchmark.ts* and *src/utils/economics.ts*.